



Open Source, Digital Sovereignty and Europe's Geospatial Future

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FOSS4G Europe 2026 – Timișoara, Romania, 30 June 2026





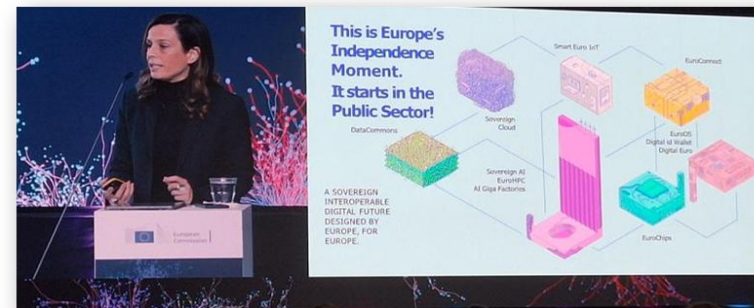
Intro – European Digital Sovereignty

European Digital Sovereignty

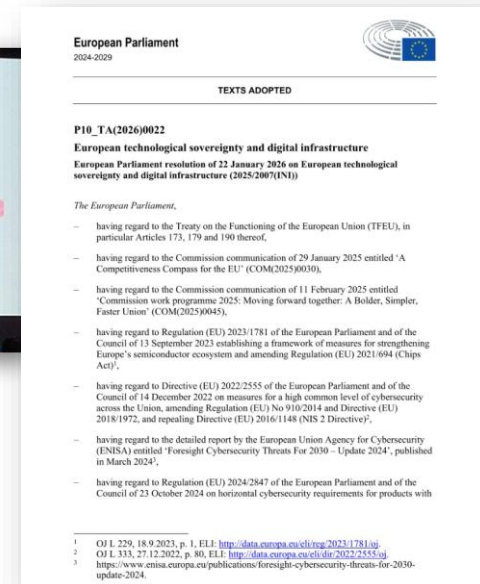
- A new **geopolitical phase** with digital technologies as **infrastructure of powers**
- **Political momentum**
 - Franco-German dialogue, Vienna Declaration on Digital Sovereignty
 - EU Parliament Resolution on Technological Sovereignty and Digital Infrastructures
 - EuroStack Report
- Risk of “**sovereign-washing**”
- Priority setting **not always clear**
- **Science for policy** can provide valuable insights



Summit on European Digital Sovereignty, Nov 2025



EuroStack report presented by Francesca Bria, Nov 2025

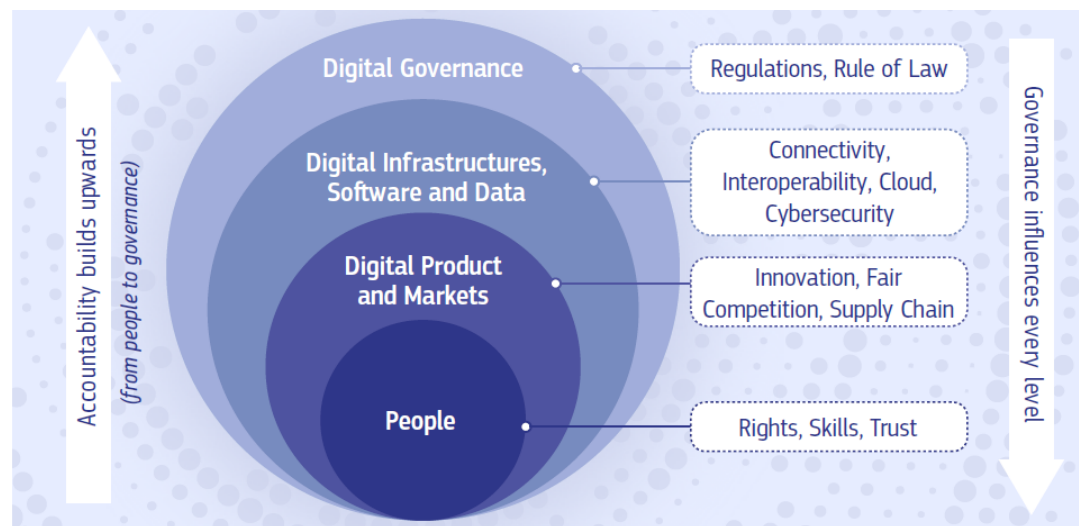


EP Resolution on Technological Sovereignty and Digital Infrastructures, Jan 2026

A Science for Policy approach to digital sovereignty



- EU digital sovereignty as the EU's **capacity of exercising its independence in the digital domain while remaining open and connected to global networks**
- Focus on the ability to **decide**, **invest** and **innovate** according to European values of democracy, openness, and the rule of law
- A **multi-layered framework** to address it in a holistic way



<https://publications.jrc.ec.europa.eu/repository/handle/JRC144908>



JRC ongoing activities on digital sovereignty



Digital Sovereignty Policy Briefs Series – Holistic & Multidisciplinary Overview



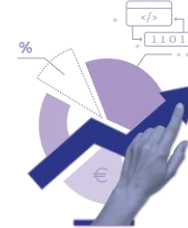
Geopolitical dimension



Environmental implications



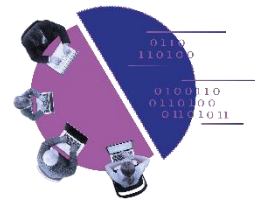
Technological implications



Economic implications



Industrial implications



Social implications

JRC Digital Sovereignty Indicators Framework

Open source assessment framework

Study on the Protection of Sensitive EU non-personal data

Digital Sovereignty Canvas + Self-assessment tool

Role plays & Policy Lab sessions

DS Seminars & Forum (Nov2026)

Research Partnerships & Collaborations



Ongoing case studies collection and experimentation
[Knowledge for Policy Platform] – LIVE SOON

SCIENCE FOR POLICY

PRACTICAL TOLLS

EVIDENCE-DRIVEN CASES



Towards an EU open source strategy

But before that

- Raise your hand if you agree with the following sentences...

I am Digitally Sovereign because I use a proprietary solution developed by a local company.



I am Digitally Sovereign because I will develop everything from scratch myself.



I am Digitally Sovereign because I have picked up an open source project, customised it internally or funded feature development and I run it myself.



Open source automatically increases digital sovereignty.



Open source shifts dependency rather than eliminating it.



Open source is a comforting illusion of control.



Call for evidence

In preparation

Call for evidence

Feedback period

06 January 2026 - 03 February 2026

Feedback: Closed

Upcoming

Commission adoption

Planned for

First quarter 2026

About this initiative

Summary	The European Open Digital Ecosystem Strategy - a strategic approach to the open source software ecosystem, highlighting the importance of open source as a crucial element for digital sovereignty, security and competitiveness - a strategic and operational framework to stimulate the reuse of open digital assets within the Commission, achieved under the 2020-2023 Commission
Topic	Digital economy and society
Type of act	Communication
Category	REFIT

Call for evidence

Feedback: Closed

Feedback period
06 January 2026 - 03 February 2026 (midnight Brussels time)

Feedback (1658)



Feedback from: Open Source Geospatial Foundation (OSGeo)

EN
(232.3 KB - PDF - 14 pages)

Download

Introduction

As a leading global community for Free and Open Source Geospatial Software, **OSGeo.org** welcomes the European Commission's Communication "**Towards European Open Digital Ecosystems**" (Ares(2026)69111).

For over two decades, the Open Source Geospatial Foundation (OSGeo) has served as an umbrella foundation for global open source geospatial software like QGIS, gvSIG, GeoServer, MapServer, GDAL, PostGIS, GeoTools, OpenLayers, GRASS, GeoNetwork, PygeoAPI, Rasdaman, Zoo and many others, providing key components to the European spatial data infrastructure. It provides public authorities, SMEs, and researchers with a sovereign, interoperable, and cost-effective alternative to proprietary software [0]. We endorse the Commission's focus on cementing digital autonomy and open collaboration. Hence, we submit this evidence to emphasise that a truly resilient European digital ecosystem requires sustained, systemic support for the underlying open source communities that maintain it. By prioritising "Open Source First" procurement and investing in the security and scalability of community-driven tools, the EU can ensure that its geospatial future is built on transparent, democratic, and truly sustainable foundations.

While open-source software is already widely used across European public administrations, widespread usage alone does not guarantee its long-term sustainability. The reliability, security, and strategic value of Digital Public Goods depend on continuous investment in maintenance, governance, and community coordination. Without deliberate policy and procurement mechanisms that distinguish between merely using open source and actively sustaining it, Europe risks building critical digital services on infrastructures whose long-term resilience is underfunded.

Despite hosting several large open source geospatial communities (e.g. QGIS, Leaflet, GeoServer, GeoNetwork), Europe's institutional infrastructure is locked into a non-EU proprietary monopoly. ESRI's dominance, fueled by free academic licensing and restrictive public procurement, creates a "sovereignty gap" where critical national data is managed via proprietary software. This concentration hinders technological autonomy and captures economic value outside the Union. To meet the goals of **Ares(2026)69111**, Europe must ensure its digital roadmap is dictated by public interest, not global commercial providers.

<https://link.europa.eu/j7Q3x4>



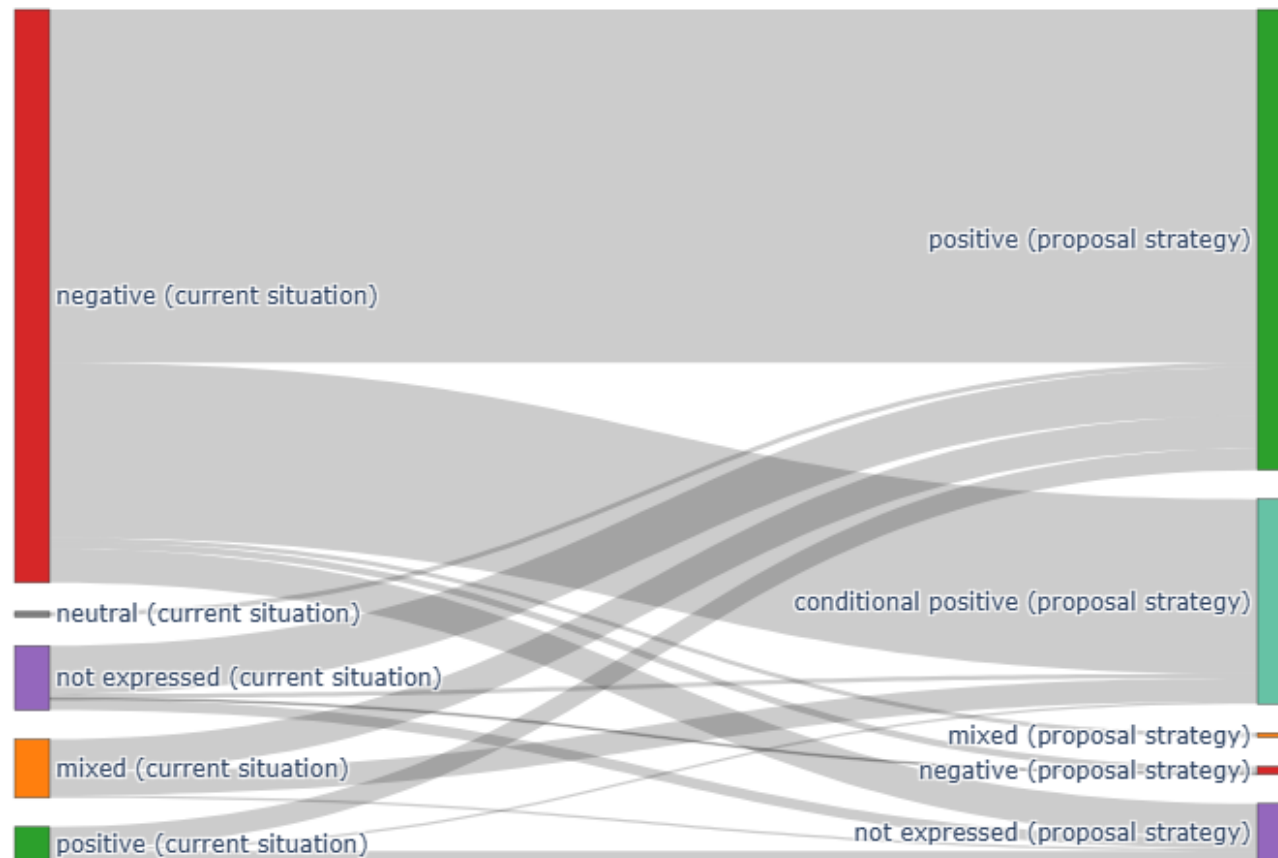
European
Commission

Analysis of feedback from the consultation

- All comments (textual submissions & metadata) extracted, translated in English and processed in GPT@JRC
- Contribution by **respondent**
 - EU citizens (>60%), businesses (~20%), NGOs (~5%), others
- Contributions by **country**
 - DE (~20%), FR (~17,5%), IT (~12,5%), NL (~10%), BE (~8%), AT (~7,5%)
- Contribution by primary **topic** (thematic tagging using Zero-Shot AI)
 - digital sovereignty and strategic autonomy (~40%), open-source software and community support (~30%)

Multi-dimensional Sentiment Analysis

- Capturing the user positions on the **current** & **future** situation (i.e. with the Strategy in place)



- More than half of submitted feedback is **negative** about the current situation, however almost all the respondents are **positive** (or conditionally positive) on the proposed strategy

SWOT analysis

- SWOT only derived from **explicit statements** in each comment

Strengths

- Enhances transparency and accountability in the public sector
- Reduces vendor lock-in and dependency
- Increases system resilience via community scrutiny
- Enables community-driven innovation
- Accelerates collaborative development
- Ensures cost efficiency compared to proprietary solutions
- Supports long-term technological sustainability
- Promotes interoperability
- Fosters skills development and capacity building

Weaknesses

- Insufficient and unstable funding for long-term sustainability
- Continued reliance on proprietary solutions in public procurement
- Limited structured support for maintenance of critical projects
- Fragmented adoption across Member States and institutions
- Weak integration of open source into procurement frameworks
- Skills gaps among public sector professionals
- Limited awareness among decision-makers
- Governance challenges in scaling community-driven projects
- Overdependence on volunteer contributions for essential infrastructure

Opportunities

- Strengthening European digital sovereignty through strategic open-source investment
- Increasing public funding and coordinated EU-level support mechanisms
- Reforming procurement rules to favor open-source adoption
- Creating shared European digital infrastructure and platforms
- Leveraging open-source to enhance cybersecurity and trust
- Supporting SMEs and startups through open digital ecosystems
- Developing structured programs for long-term project sustainability

Threats

- Continued dependence on foreign proprietary vendors
- Underfunding of critical open-source infrastructure
- Security risks due to insufficient maintenance and governance
- Regulatory fragmentation across Member States
- Market dominance by large non-European technology providers
- Insufficient adoption reducing European competitiveness
- Loss of community engagement without institutional support
- Strategic vulnerability in critical digital infrastructure



The EU Open Source Strategy

European Tech Sovereignty Package

The Technological Sovereignty Package will strengthen Europe's competitiveness, strategic autonomy and geoeconomic position.

WHAT IT MEANS



Boosting homegrown industrial capacity and autonomy across the supply chain.



Increasing choice for end-users.



Securing the supply of digital technologies.



Developing, controlling and scaling critical technologies, infrastructure, services and data.



Setting the standards for key strategic technologies.

THE PACKAGE INCLUDES

CHIPS ACT 2.0

CLOUD AND
AI DEVELOPMENT ACT

EU
OPEN SOURCE
STRATEGY

STRATEGIC ROADMAP
FOR DIGITALISATION
AND AI IN ENERGY



Free and open-source software is **recognised as a strategic asset** for Europe and as critical digital infrastructure!

The EU Open Source Strategy

Four key objectives of the EU Open Source Strategy



Open Source for tech sovereignty

Boost the development and adoption of open source alternatives in line with EU values.



A vibrant Open Source ecosystem

Developed through businesses, communities, governance, security and maintenance, skills.



Open Source in public administration

- Provide guidelines and best practices for procurement.
- Strengthen the Commission's own use of open source.



Reinforced standards and international outreach

- EU Tech Business Offer: promoting open source solutions internationally.
- Better integration of open source into standardisation processes.

Objective 2

Vibrant Open Source Ecosystem



Businesses

- Support startups via accelerators, legal/licensing help, training, and procurement opportunities.

Stewardship

- Develop a stewardship toolkit and support EU-based steward organisations for strategic assets.

Security & Maintenance

- Create an Open Source Maintenance Instrument, critical dependency mapping, and mirroring capabilities.

Skills

- Invest in skills for schools, universities, civil servants, and learners.

The EU Open Source Strategy & Business



| **European Alternatives**

Developing European open source solutions supports greater choice for businesses, public authorities and citizens while respecting EU values. There are already over 500 for profit open source companies in Europe working in key sectors including cloud, software defined industrial systems, cybersecurity and data.



| **Greater competitiveness**

EU companies, especially SMEs, can benefit from increased choice, accelerate innovation and develop new business models by integrating open source in their solutions. Businesses in traditional sectors can use open source to speed up their digital transition and innovate collaboratively.

2027+ - Open Source Strategy creates Opportunity

- Upcoming open-source maintenance instrument, support the maintenance and security upkeep
- Digital Commons EDIC and Memberstate activities (Sovereign Tech Fund)
- Open Source Policy Labs (to test and evaluate FOSS solutions for public administration)
- Common security guidelines for Commission repositories: monitoring, vulnerabilities, licence compliance, and dependency risk.



Future Opportunities

NOW - A services and business incubator for geospatial open-source developments



CONTEXT

Open-source software at the core of geospatial services -> **need for long-term economic sustainability**



EXPECTED OUTCOMES

- A **business incubation and support hub** for the EU geospatial open-source community
- Develop **sustainability & sovereignty strategies** for geospatial digital commons
- **Capacity-building**: business mentoring, IP guidance, investor matchmaking



PREPARE TO ENGAGE

- Discuss, implement and update lightweight **governance mechanisms** for your FOSS project
- Create a **development roadmap** for future maintenance needs, and updated functionality
- **Map entrepreneurial interest** in your project, including NGO, Social Entrepreneurship and Start-ups



PLEASE NOTE

- Collaborative Support Action
- Financial support to third parties of **2.7 -3.6 Million**
- **Additional: Open Internet Stack FSTP programme**

**PROJECT START IS EXPECTED IN
Q4 2026 or Q1 2027**

NOW - Become an Independent Expert

WORK AS AN EVALUATOR



- ✓ Academia
- ✓ Private sector / innovation agencies
- ✓ Local / national authorities / agencies

Call for experts:

<https://ec.europa.eu/info/funding-tenders/opportunities/portal/screen/work-as-an-expert>

REA is looking for experts to help evaluate funding proposals and review pioneering projects. Subject-matter experts from various fields – from public authorities and civil society organisations to universities, businesses and research organisations – are invited to register in the [European Commission's experts database](#).



2027 - Open source assessment framework

- Voluntary self-assessment framework for open source software projects
 - measuring **quality** – maturity, popularity, development/maintenance team size, transparency of governance, quality of code/documentation, use of standards, licensing, EU presence
- Conceived as a **co-creation** exercise with open source communities
 - taking into consideration other existing assessment frameworks
 - open source communities (incl. OSGeo) to be involved
 - open source project itself

Thank you



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